

Using local ecdata at: /home/kepler/ecdata
Using local lmfdb data at: /home/kepler/lmfdb

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UCF LOCAL DB Tool — Full ACSC Projection

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Starting parallel processing with 6 cores...

> Processing 'Virgo' (a=-1706, b=6320)... > Processing 'Coma' (a=-10141, b=9980)... >
Processing 'Hercules' (a=-15796, b=8500)... > Processing 'Perseus' (a=-7456, b=11500)... >
Processing 'Centaurus' (a=-5371, b=7500)... > Processing 'Fornax' (a=-1959, b=3200)...

Creating Sage model for a=-1706, b=6320...
Creating Sage model for a=-15796, b=8500... Creating Sage model for a=-10141, b=9980...
Creating Sage model for a=-7456, b=11500... Creating Sage model for a=-1959, b=3200...
Creating Sage model for a=-5371, b=7500...

Minimal model: (0, 1, 0, -963825, -364520177) Minimal model: (0, -1, 0, -18519145,
-30668465975)

Computing rank with high two_descent limit...

Computing rank with high two_descent limit...
Minimal model: (0, -1, 0, -83162705, -291876936975) Minimal model: (0, -1, 0, -9608380,
-11460457100) Minimal model: (0, -1, 0, -34269980, -77206636000) Minimal model: (1, -1,
1, -79752, -8648821)

Computing rank with high two_descent limit... Computing rank with high two_descent limit...
Computing rank with high two_descent limit... Computing rank with high two_descent limit...

> Processing 'Shapley' (a=-20535, b=18000)... > Processing 'Horologium' (a=-22115,
b=12000)...

Creating Sage model for $a=-20535$, $b=18000...$ Creating Sage model for $a=-22115$,
 $b=12000...$

Minimal model: $(1, -1, 1, -8784005, -10018247628)$ Minimal model: $(1, 0, 1, -10188276,$
 $-12517816802)$

Computing rank with high two_descent limit... Computing rank with high two_descent limit...

1 points of order 2:
[-569:0:1]

Using 2-isogenous curve $[0, 3412, 0, 2885156, 0]$ (minimal model $[0, 1, 0, -995425, -339372897]$)

1 points of order 2:
[-2485:0:1]

Using 2-isogenous curve $[0, 14912, 0, 55545936, 0]$ (minimal model
 $[0, -1, 0, -1161040, -475778138]$)

1 points of order 2:
[-3380:0:1]

1 points of order 2:
[-1790:0:1]

Using 2-isogenous curve $[0, 20282, 0, 102799961, 0]$ (minimal model
 $[0, -1, 0, -34319880, -76970469280]$)

Using 2-isogenous curve $[0, 10742, 0, 28817641, 0]$ (minimal model
 $[0, -1, 0, -9645880, -11366452100]$)

1 points of order 2:
[-5265:0:1]

Using 2-isogenous curve $[0, 31592, 0, 249479616, 0]$ (minimal model
 $[0, -1, 0, -5200325, -4554381723]$)

1 points of order 2:
[-163:81:1]

Using 2-isogenous curve $[0, 3918, 0, 3824881, 0]$ (minimal model $[1, -1, 1, -80752, -8420021]$)

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 1

$$\text{rk}(S^{\{\phi\}}(E')) = 1$$

$$\text{rk}(S^{\{\phi'\}}(E)) = 2$$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 1

$$\text{rk}(\phi'(S^2(E))) = 1$$

$$\text{rk}(\phi(S^2(E))) = 2$$

$$\text{rk}(S^2(E)) = 2$$

$$\text{rk}(S^2(E')) = 2$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

 $(c,d) = (-7456, 11500)$

$$(c',d') = (14912, 55545936)$$

This component of the rank is 0

2. $E'(Q)/\phi'(E(Q))$

First stage (no second descent yet)...

$$(-51, 0, 14912, 0, -1089136) \text{-----}$$

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 0

$$\text{rk}(S^{\{\phi\}}(E')) = 1$$

$$\text{rk}(S^{\{\phi'\}}(E)) = 1$$

Second step, determining 2nd descent Selmer groups

...skipping since we already know rank=0

After second local descent, rank bound = 0

$$\text{rk}(\phi'(S^2(E))) = 1$$

$$\text{rk}(\phi(S^2(E))) = 1$$

$$\text{rk}(S^2(E)) = 1$$

$$\text{rk}(S^2(E')) = 1$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$(c,d) = (-1959, 3200)$

$(c',d') = (3918, 3824881)$

This component of the rank is 0

2. $E'(Q)/\phi'(E(Q))$

This component of the rank is 0

Summary of results:

$\text{rank}(E) = 0$

$\#E(Q)/2E(Q) = 2$

Information on $\text{III}(E/Q)$:

$\#\text{III}(E/Q)[\phi'] = 1$

$\#\text{III}(E/Q)[2] = 1$

Information on $\text{III}(E'/Q)$:

$\#\phi'(\text{III}(E'/Q)[2]) = 1$

$\#\text{III}(E'/Q)[\phi] = 1$

$\#\text{III}(E'/Q)[2] = 1$

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 2

$\text{rk}(S^{\{\phi\}}(E')) = 1$

$\text{rk}(S^{\{\phi\}}(E)) = 3$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 0

$\text{rk}(\phi'(S^{\{2\}}(E))) = 1$

$\text{rk}(\phi(S^{\{2\}}(E))) = 1$

$\text{rk}(S^{\{2\}}(E)) = 3$

$\text{rk}(S^{\{2\}}(E')) = 1$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$(c,d) = (-15796,8500)$
 $(c',d') = (31592,249479616)$
 This component of the rank is 0

2. $E'(Q)/\phi'(E(Q))$

This component of the rank is 0

Summary of results:

$$\begin{aligned}\text{rank}(E) &= 0 \\ \#E(Q)/2E(Q) &= 2\end{aligned}$$

Information on $\text{III}(E/Q)$:

$$\begin{aligned}\#\text{III}(E/Q)[\phi'] &\leq 4 \\ \#\text{III}(E/Q)[2] &= 4\end{aligned}$$

Information on $\text{III}(E'/Q)$:

$$\begin{aligned}\#\phi'(\text{III}(E/Q)[2]) &= 1 \\ \#\text{III}(E'/Q)[\phi] &= 1 \\ \#\text{III}(E'/Q)[2] &= 1\end{aligned}$$

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 2

$$\begin{aligned}\text{rk}(S^{\{\phi\}}(E')) &= 2 \\ \text{rk}(S^{\{\phi'\}}(E)) &= 2\end{aligned}$$

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 1

$$\begin{aligned}\text{rk}(S^{\{\phi\}}(E')) &= 2 \\ \text{rk}(S^{\{\phi'\}}(E)) &= 1\end{aligned}$$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 1

$$\begin{aligned}\text{rk}(\phi'(S^{\{2\}}(E))) &= 2 \\ \text{rk}(\phi(S^{\{2\}}(E))) &= 1\end{aligned}$$

$$\text{rk}(S^2(E)) = 2$$

$$\text{rk}(S^2(E')) = 2$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$$(c,d) = (-5371, 7500)$$

$$(c',d') = (10742, 28817641)$$

First stage (no second descent yet)...

$$(5, 0, -5371, 0, 1500) \text{-----}$$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 2

$$\text{rk}(\phi'(S^2(E))) = 2$$

$$\text{rk}(\phi(S^2(E'))) = 2$$

$$\text{rk}(S^2(E)) = 3$$

$$\text{rk}(S^2(E')) = 3$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$$(c,d) = (-1706, 6320)$$

$$(c',d') = (3412, 2885156)$$

First stage (no second descent yet)...

$$(2, 0, -1706, 0, 3160) \text{-----}$$

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 2

$$\text{rk}(S^{\phi}(E')) = 2$$

$$\text{rk}(S^{\phi'}(E)) = 2$$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 2

$$\text{rk}(\phi'(S^2(E))) = 2$$

$$\text{rk}(\phi(S^2(E'))) = 2$$

$$\text{rk}(S^2(E)) = 3$$

$$\text{rk}(S^2(E')) = 3$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

(c,d) = (-10141,9980)

(c',d')=(20282,102799961)

First stage (no second descent yet)...

(5,0,-10141,0,1996) Processing points found during 2-descent...done:

now regulator = 1

Processing points found during 2-descent...done:

now regulator = 1

: (x:y:z) = (2812:1616780:233)

Curve E' Point [-93962968752:-231865653360:12649337], height = 13.96405844

: no rational point found (hlim=8)

(499,0,-10141,0,20): (x:y:z) = (10:1280:19)

Curve E Point [9500:64000:6859], height = 6.639234854

After first global descent, this component of the rank = 2

2. $E'(Q)/\phi'(E(Q))$

This component of the rank is 0

Summary of results:

rank(E) = 1

$\#E(Q)/2E(Q) = 4$

Information on $\text{III}(E/Q)$:

$\#\text{III}(E/Q)[\phi'] = 1$

$\#\text{III}(E/Q)[2] = 1$

Information on $\text{III}(E'/Q)$:

$\#\phi'(\text{III}(E'/Q)[2]) = 1$

$\#\text{III}(E'/Q)[\phi] = 1$

$\#\text{III}(E'/Q)[2] = 1$

✓ Completed Hercules

✓ Completed Fornax

: (x:y:z) = (51:3016:1)

Curve E Point [5202:307632:1], height = 7.863294224

After first global descent, this component of the rank = 2

2. $E'(Q)/\phi'(E(Q))$

First stage (no second descent yet)...

(-71,0,3412,0,-40636): no rational point found (hlim=8)

After first global descent, this component of the rank

has lower bound 0

and upper bound 1

(difference = 1)

Second descent will attempt to reduce this

Second stage (using second descent)...

d1=5: Curve E Point

[53520863965793950:534929566532567525:35157964649371192], height = 27.92811688

After first global descent, this component of the rank = 2

Summary of results:

rank(E) = 1

#E(Q)/2E(Q) = 4

Information on III(E/Q):

#III(E/Q)[phi] = 1

#III(E/Q)[2] = 1

Information on III(E'/Q):

#phi'(III(E/Q)[2]) = 1

#III(E'/Q)[phi] = 1

#III(E'/Q)[2] = 1

1 points of order 2:

[-1843:921:1]

Using 2-isogenous curve [0,44230,0,489025225,0] (minimal model

[1,0,1,-10192026,-12508141802])

1 points of order 2:

[-1711:855:1]

Using 2-isogenous curve [0,41070,0,421614225,0] (minimal model

[1,-1,1,-8789630,-10004770128])

Second descent inconclusive for d1=5: ELS descendents exist but no rational point found

d1=499: -----

List of points on E = [0,-1,0,-9608380,-11460457100]:

I. Points on $E \bmod \phi(E')$

Point $[-12268110:64000:6859]$, height = 6.639234854

II. Points on $\phi(E') \bmod 2E$

--none (modulo torsion).

Computing full set of 2 coset representatives for
 $2E(Q)$ in $E(Q)$ (modulo torsion), and sorting into height order....done.

Second descent inconclusive for $d1=499$: ELS descendents exist but no rational point found

After second global descent, this component of the rank
has lower bound 0
and upper bound 1
(difference = 1)

2. $E'(Q)/\phi'(E(Q))$

First stage (no second descent yet)...

$(-11,0,20282,0,-9345451)$: no rational point found (hlim=8)

$(-9345451,0,20282,0,-11)$: $(x:y:z) = (5:17:1)$

Curve E' Point $[-1775:-6035:1]$, height = 4.490028899

List of points on $E = [0,-1,0,-18519145,-30668465975]$:

I. Points on $E \bmod \phi(E')$

--none (modulo torsion).

II. Points on $\phi(E') \bmod 2E$

Point $[-87314021289721618170:534929566532567525:35157964649371192]$, height = 27.92811688

Computing full set of 2 coset representatives for
 $2E(Q)$ in $E(Q)$ (modulo torsion), and sorting into height order....done.

: no rational point found (hlim=8)

After first global descent, this component of the rank
has lower bound 0
and upper bound 1

(difference = 1)

Second descent will attempt to reduce this

Second stage (using second descent)...

d1=-11:

Second descent inconclusive for d1=-11: ELS descendents exist but no rational point found

d1=-9345451: Curve E Point [2890:-63563:1000], height = 8.980057799

After first global descent, this component of the rank = 2

Summary of results:

rank(E) = 2

#E(Q)/2E(Q) = 8

Information on III(E/Q):

#III(E/Q)[phi'] = 1

#III(E/Q)[2] = 1

Information on III(E'/Q):

#phi'(III(E/Q)[2]) = 1

#III(E'/Q)[phi] = 1

#III(E'/Q)[2] = 1

Second descent inconclusive for d1=-9345451: ELS descendents exist but no rational point found

After second global descent, this component of the rank

has lower bound 0

and upper bound 1

(difference = 1)

Summary of results:

0 <= rank(E) <= 2

#E(Q)/2E(Q) >= 2

Information on III(E/Q):

#III(E/Q)[phi'] <= 2

#III(E/Q)[2] <= 4

Information on III(E'/Q):

$\#\phi'(\text{III}(E/Q)[2]) \leq 2$
 $\#\text{III}(E'/Q)[\phi] \leq 2$
 $\#\text{III}(E'/Q)[2] \leq 4$

Searching for points (bound = 8)...done:
 found points which generate a subgroup of rank 0
 and regulator 1

Processing points found during 2-descent.....

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 1

$\text{rk}(S^{\{\phi\}}(E')) = 1$
 $\text{rk}(S^{\{\phi\}}(E)) = 2$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 1

$\text{rk}(\phi'(S^{\{2\}}(E))) = 1$
 $\text{rk}(\phi(S^{\{2\}}(E))) = 2$
 $\text{rk}(S^{\{2\}}(E)) = 2$
 $\text{rk}(S^{\{2\}}(E')) = 2$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$(c,d) = (-20535, 18000)$
 $(c',d') = (41070, 421614225)$
 This component of the rank is 0

2. $E'(Q)/\phi'(E(Q))$

First stage (no second descent yet)...

$(-919, 0, 41070, 0, -458775)$

List of points on $E = [0, 1, 0, -963825, -364520177]$:

I. Points on $E \bmod \phi(E')$

Point $[4633:307632:1]$, height = 7.863294224

II. Points on $\phi(E') \bmod 2E$

Point $[-566110:-63563:1000]$, height = 8.980057799

Computing full set of 4 coset representatives for $2E(Q)$ in $E(Q)$ (modulo torsion), and sorting into height order....Searching for points (bound = 8).....

First step, determining 1st descent Selmer groups

After first local descent, rank bound = 1

$$\text{rk}(S^{\{\phi\}}(E')) = 2$$

$$\text{rk}(S^{\{\phi\}}(E)) = 1$$

Second step, determining 2nd descent Selmer groups

After second local descent, rank bound = 1

$$\text{rk}(\phi'(S^{\{2\}}(E))) = 2$$

$$\text{rk}(\phi(S^{\{2\}}(E))) = 1$$

$$\text{rk}(S^{\{2\}}(E)) = 2$$

$$\text{rk}(S^{\{2\}}(E')) = 2$$

Third step, determining $E(Q)/\phi(E'(Q))$ and $E'(Q)/\phi'(E(Q))$

1. $E(Q)/\phi(E'(Q))$

$$(c,d) = (-22115, 12000)$$

$$(c',d') = (44230, 489025225)$$

First stage (no second descent yet)...

(3,0,-22115,0,4000)done:

2-descent increases rank to 1, now regulator = 6.639234854

Saturating (with bound = -1)...done:

found points which generate a subgroup of rank 0
and regulator 1

Processing points found during 2-descent...done.

done:

2-descent increases rank to 1, now regulator = 27.92811688

Saturating (with bound = -1)...done:

points were already saturated.

: no rational point found (hlim=8)

(-2039,0,41070,0,-206775): no rational point found (hlim=8)

(10,0,-22115,0,1200): no rational point found (hlim=8)

After first global descent, this component of the rank

has lower bound 0

and upper bound 1

(difference = 1)

Second descent will attempt to reduce this

Second stage (using second descent)...

d1=3: : no rational point found (hlim=8)

After first global descent, this component of the rank

has lower bound 0

and upper bound 1

(difference = 1)

Second descent will attempt to reduce this

Second stage (using second descent)...

d1=-919:

Second descent inconclusive for d1=3: ELS descendents exist but no rational point found

d1=10: Searching for points (bound = 8)...

Second descent inconclusive for d1=-919: ELS descendents exist but no rational point found

d1=-2039:

Second descent inconclusive for d1=10: ELS descendents exist but no rational point found

After second global descent, this component of the rank

has lower bound 0

and upper bound 1

(difference = 1)

2. $E'(Q)/\phi'(E(Q))$

This component of the rank is 0

Summary of results:

$0 \leq \text{rank}(E) \leq 1$

$\#E(Q)/2E(Q) \geq 2$

Information on $\text{III}(E/Q)$:

$\#\text{III}(E/Q)[\phi'] = 1$

$\#\text{III}(E/Q)[2] \leq 2$

Information on $\text{III}(E'/Q)$:

$\#\phi'(\text{III}(E'/Q)[2]) \leq 2$

$\#\text{III}(E'/Q)[\phi] \leq 2$

$\#\text{III}(E'/Q)[2] \leq 2$

Second descent inconclusive for d1=-2039: ELS descendents exist but no rational point found

After second global descent, this component of the rank
has lower bound 0
and upper bound 1
(difference = 1)

Summary of results:

$0 \leq \text{rank}(E) \leq 1$
 $\#E(Q)/2E(Q) \geq 2$

Information on $\text{III}(E/Q)$:

$\#\text{III}(E/Q)[\text{phi}'] \leq 2$
 $\#\text{III}(E/Q)[2] \leq 2$

Information on $\text{III}(E'/Q)$:

$\#\text{phi}'(\text{III}(E'/Q)[2]) = 1$
 $\#\text{III}(E'/Q)[\text{phi}] = 1$
 $\#\text{III}(E'/Q)[2] \leq 2$

done:

found points which generate a subgroup of rank 0
and regulator 1

Processing points found during 2-descent...done:

points were already saturated.

- ✓ Completed Centaurus
- ✓ Completed Perseus
- ✓ Completed Shapley

done:

2-descent increases rank to 2, now regulator = 56.5675374

Saturating (with bound = -1)...done:

points were already saturated.

- ✓ Completed Virgo

Sage Error: Unable to compute the rank, hence generators, with certainty (lower bound=0, generators found=()). This could be because $\text{Sha}(E/Q)[2]$ is nontrivial.

Try increasing descent_second_limit then trying this command again.

- ✓ Completed Horologium
- ✓ Completed Coma

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FINAL RESULTS WITH FULL PROJECTION

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Cluster	r (Mly)	rho	Derived a	Derived b	Minimal ainv	Conductor	Rank	Local
Label	projected_omega	h_eff_factor	comoving_volume	scaled_regulator	entropy_proxy	cohomology_class	cosmo_scale	note
Virgo	54	6320	-1706	6320	(0, 1, 0, -963825, -364520177)	1.823419e+10	2	
Unknown	54000000.0	1.017404	3.235314e+08	12521.758187	35.150627	12.631284	2.241420e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Coma	321	9980	-10141	9980	(0, -1, 0, -34269980, -77206636000)	4.103774e+12		
Unknown	54000000.0	1.025079	5.115630e+07	632.455532	39.637561	2.095399	5.473343e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Perseus	236	11500	-7456	11500	(0, -1, 0, -18519145, -30668465975)	2.555113e+10		
Unknown	54000000.0	1.023747	1.312817e+09	4415.823004	39.305514	4.804402	4.692772e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Centaurus	170	7500	-5371	7500	(0, -1, 0, -9608380, -11460457100)	3.458117e+09		
Unknown	54000000.0	1.022330	2.247995e+08	1049.755203	37.794404	5.384430	3.982784e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Fornax	62	3200	-1959	3200	(1, -1, 1, -79752, -8648821)	3.824881e+07	0	
Unknown	54000000.0	1.017993	2.467811e+06	632.455532	25.753673	2.902147	1.202151e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Hercules	500	8500	-15796	8500	(0, -1, 0, -83162705, -291876936975)	2.120577e+10		
Unknown	54000000.0	1.026998	7.969048e+07	632.455532	40.203119	1.720974	6.831347e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Shapley	650	18000	-20535	18000	(1, -1, 1, -8784005, -10018247628)	4.216142e+08		
Unknown	54000000.0	1.028136	2.253691e+09	11006.921033	33.910678	2.559770	3.894461e+07	
Full ACSC Φ + ECC projection (local DB, high descent)								
Success								
Horologium	700	12000	-22115	12000	N/A	NaN	None	---
NaN	NaN	NaN	NaN	NaN	NaN	NaN	Sage Error:	
Unable to compute the rank, hence generators, with certainty (lower bound=0, generators found=()). This could be because Sha(E/Q)[2] is nontrivial. Try increasing								

descent_second_limit then trying this command again. Sage Error: Unable to compute the rank, hence generators, with certainty (lower bound=0, generators found=()). This could be because $\text{Sha}(E/Q)[2]$ is nontrivial. Try increasing descent_second_limit then trying this command again.

Results saved to 'ucf_local_high_descent_acsc_projection.csv'
Execution complete. Ready for STAR.ipynb import.